

Home SOC Lab: Threat Detection & Incident Response

A Hands-On Security Operations Project

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1. Executive Summary

This project involved building a home Security Operations Center (SOC) lab to gain hands-on experience in threat detection, log analysis, and incident response. A simulated attack environment was created using VirtualBox, with a Windows 10 victim machine and Kali Linux attacker machine on an isolated network. The Windows machine was instrumented with Sysmon for detailed endpoint logging, and logs were forwarded to Splunk for centralized analysis.

Using Atomic Red Team, real-world MITRE ATT&CK techniques were simulated, including PowerShell-based execution and credential access behavior resembling Mimikatz. A detection search was built in Splunk and validated, then converted into an automated alert that successfully triggered on live simulated attack activity — demonstrating an end-to-end detection pipeline similar to those used in real SOC environments.

2. Objective

The goal of this project was to:

- Build a functioning SOC lab environment from scratch
- Gain practical experience with a SIEM (Splunk)
- Simulate real attack techniques mapped to MITRE ATT&CK
- Detect those techniques through log analysis
- Build a working, automated detection alert
- Document the process the way a SOC analyst would document a real incident

3. Lab Architecture

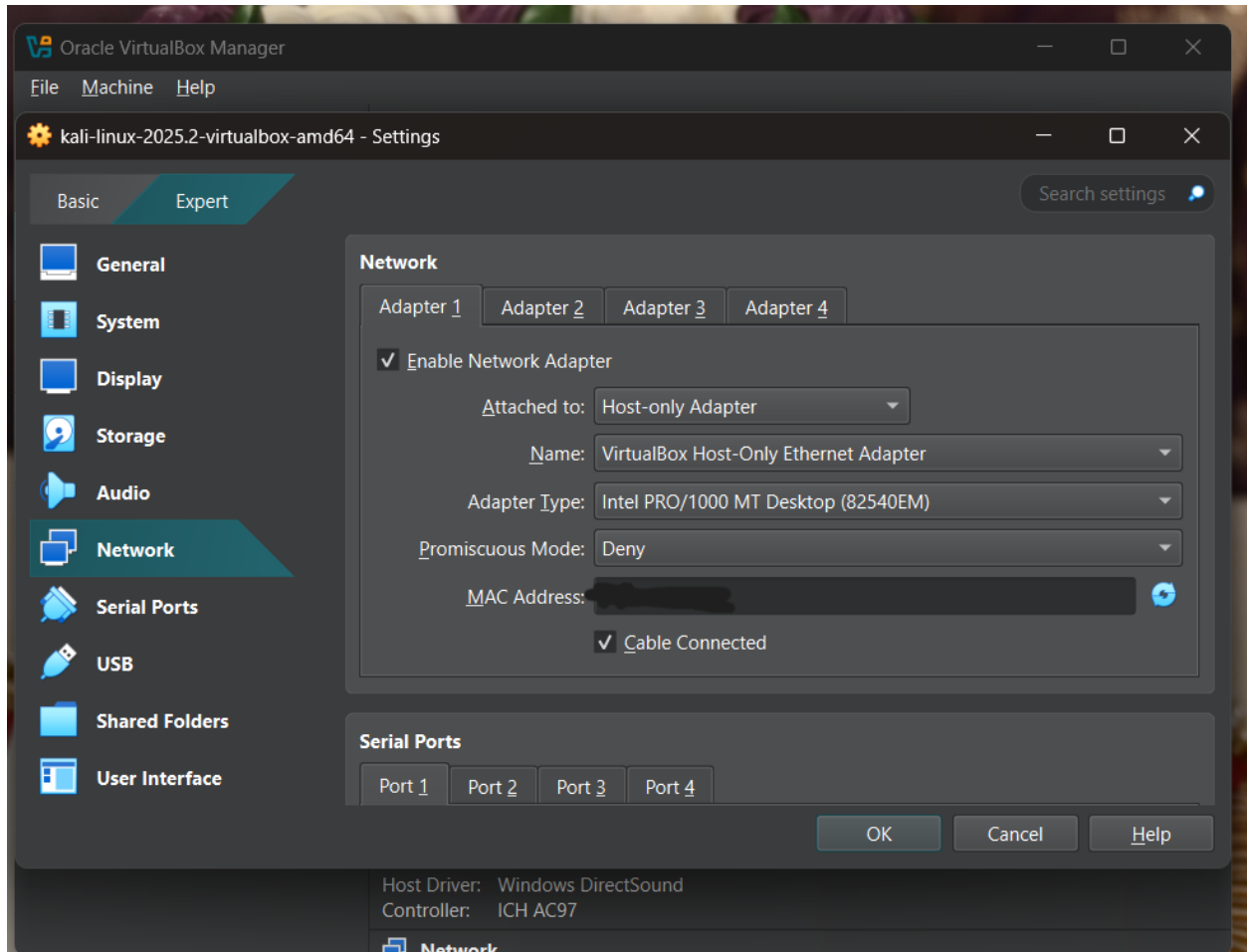
Components used:

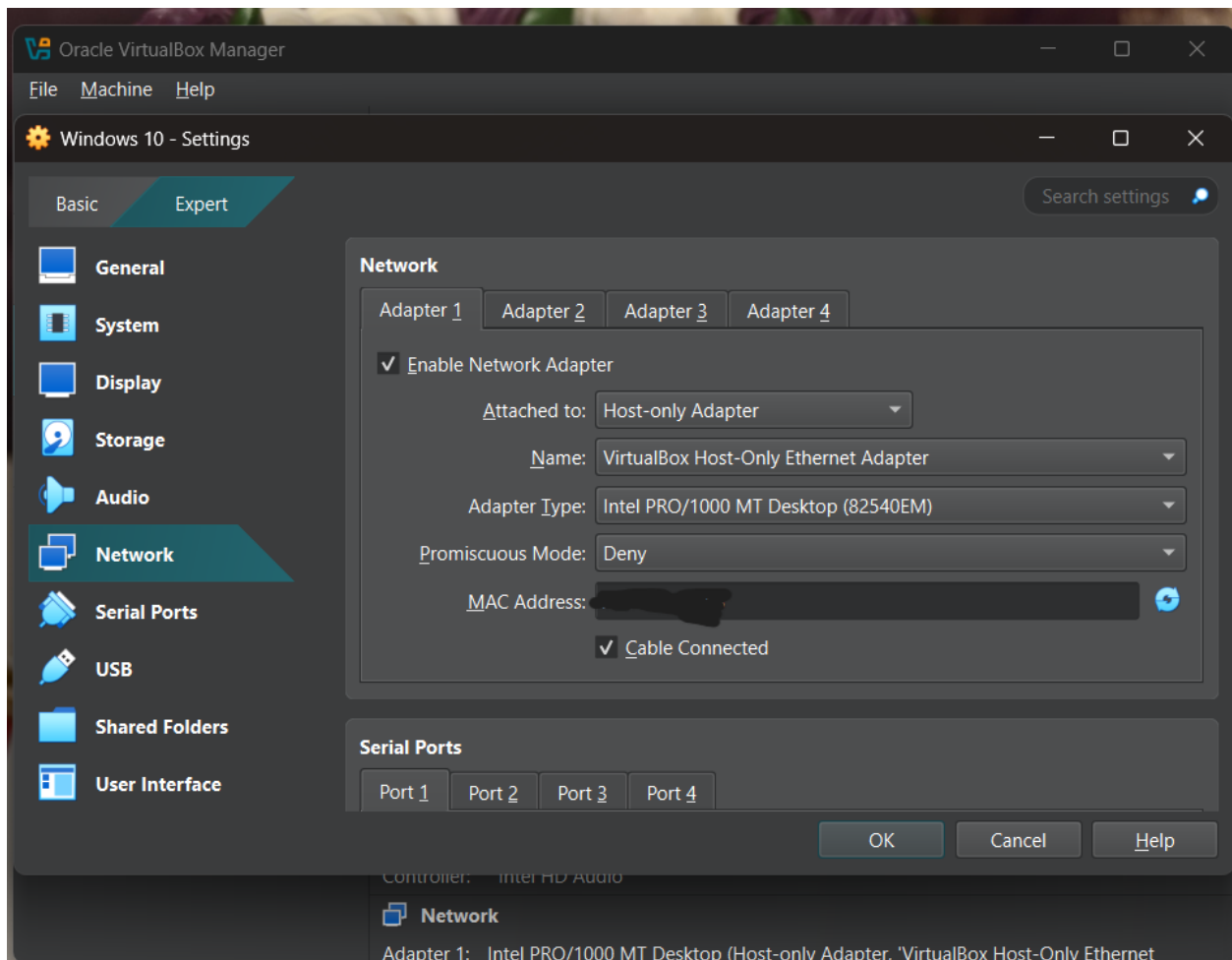
- Windows 11 (host machine) — running Splunk Enterprise (SIEM)
- Windows 10 (VirtualBox VM) — victim machine, instrumented with Sysmon
- Kali Linux (VirtualBox VM) — attacker machine
- Network: VirtualBox Host-Only Adapter connecting all machines
- Splunk Universal Forwarder — installed on Windows 10 VM to ship logs to Splunk
- Atomic Red Team — used to simulate MITRE ATT&CK techniques

4. Setup Process

4.1 Network Configuration

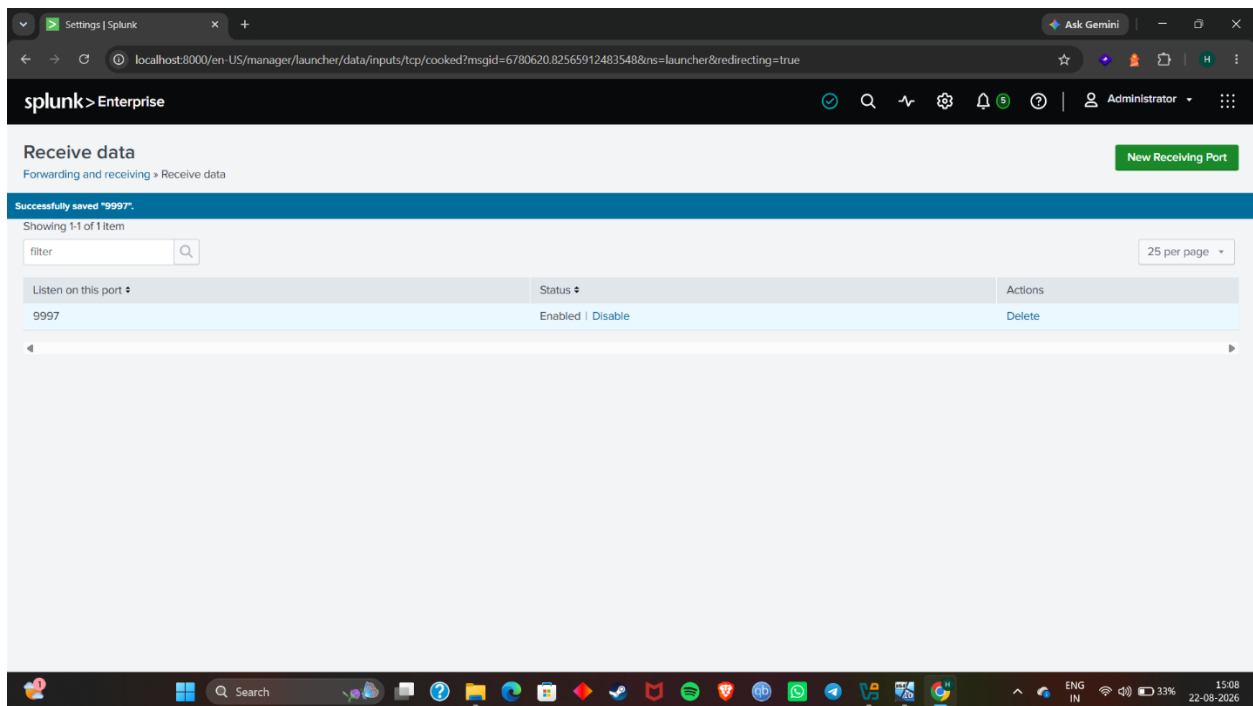
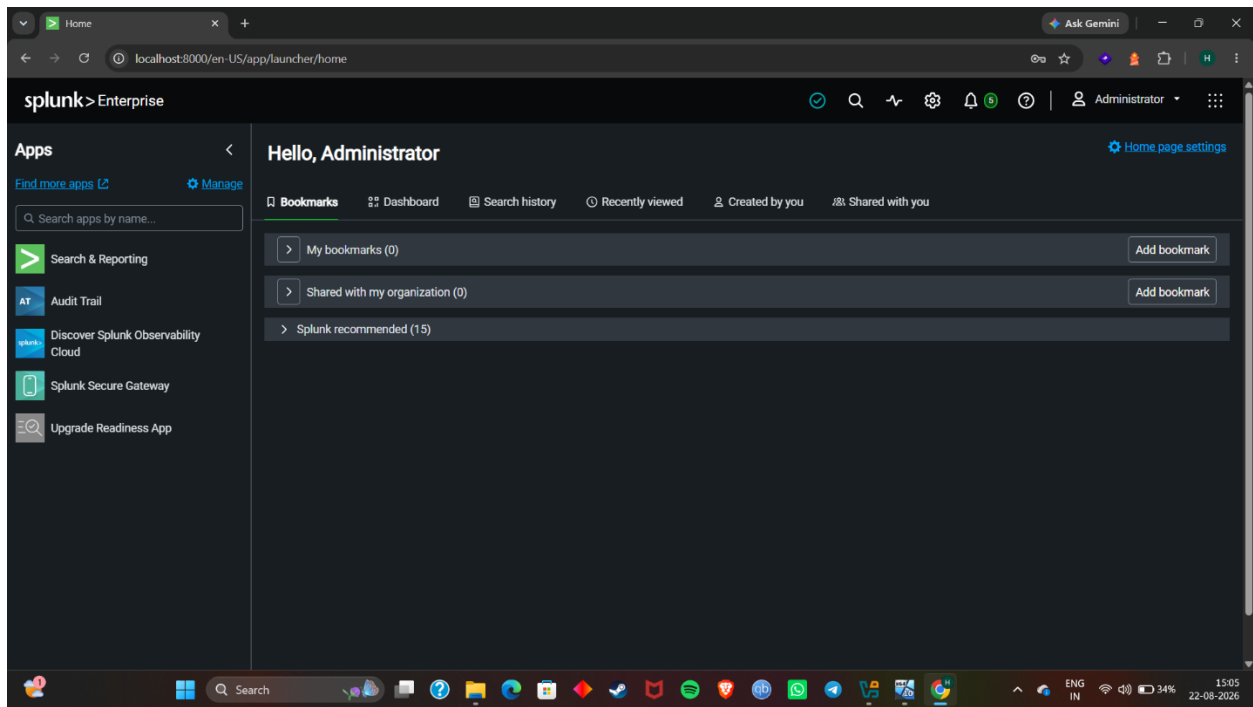
Both VMs were configured to use a Host-Only Adapter, allowing them to communicate with each other and the host machine while remaining isolated from the internet on that interface. A second NAT adapter was added to the Windows 10 VM to allow software downloads during setup.





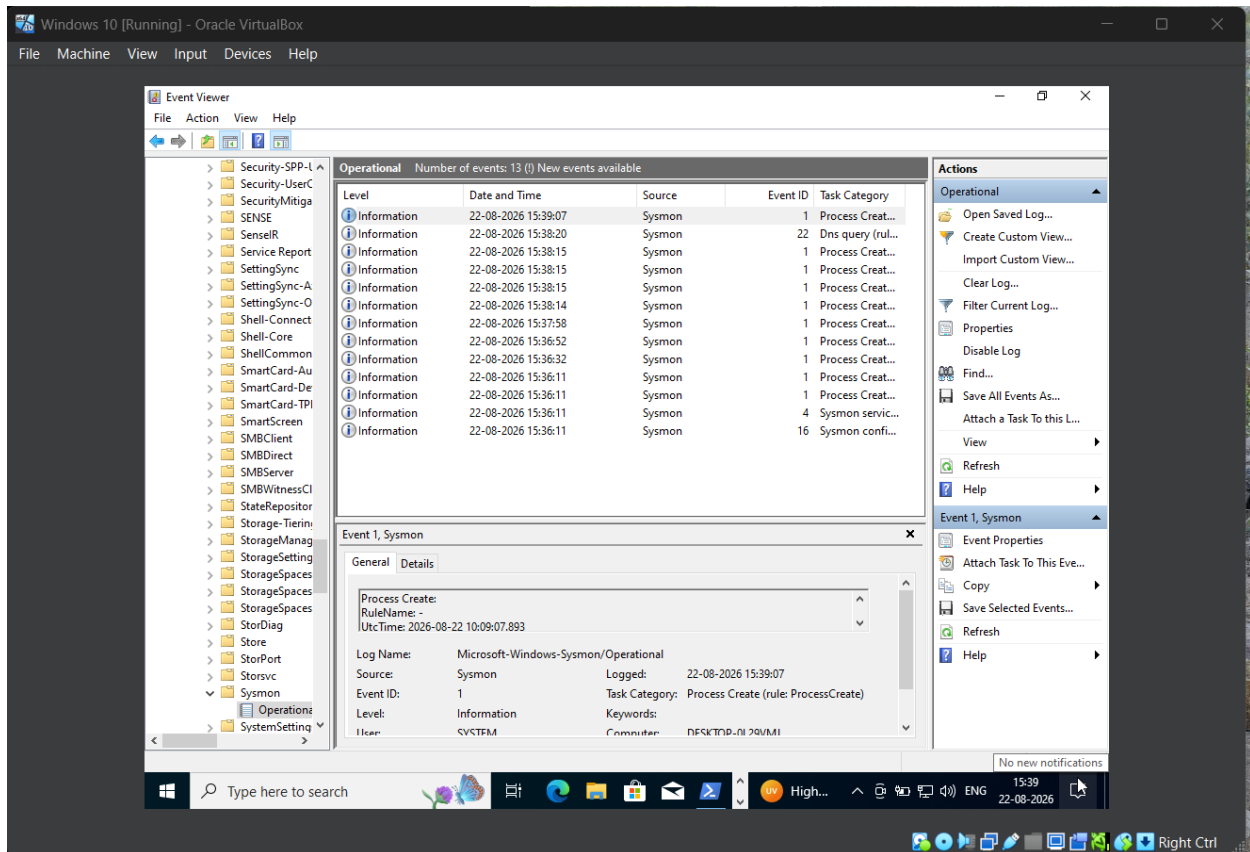
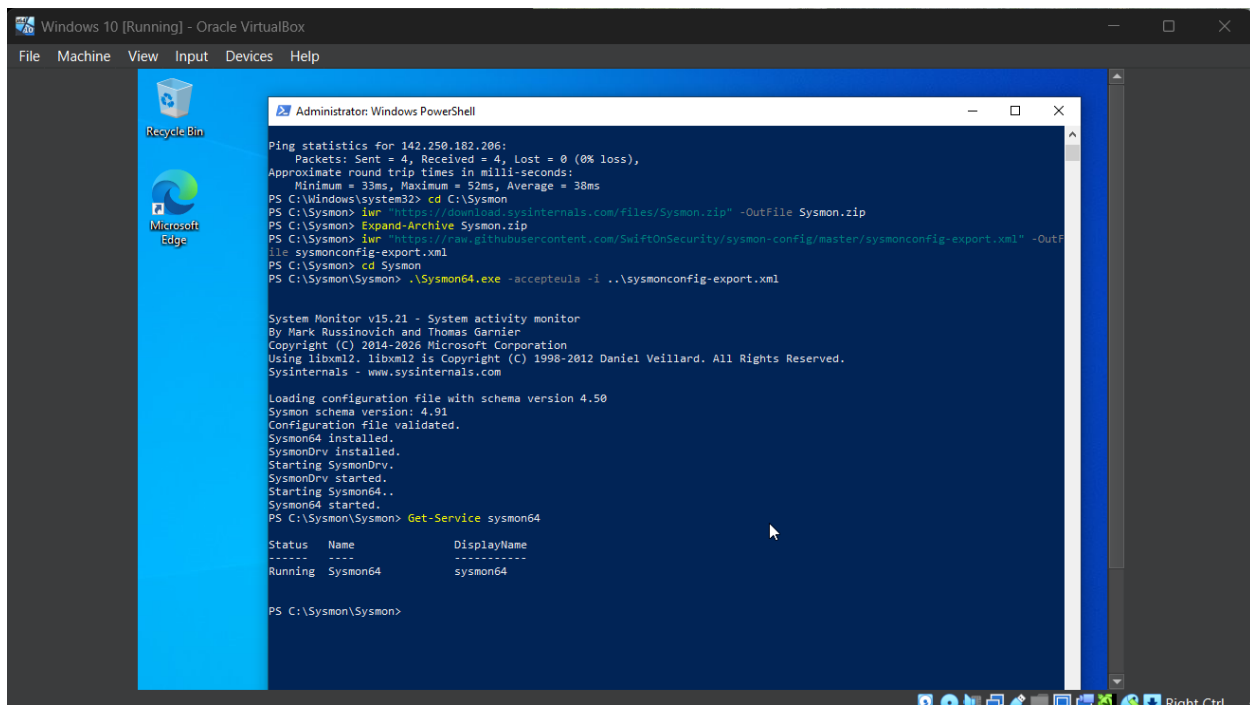
4.2 Splunk Installation

Splunk Enterprise (Free tier) was installed on the Windows 11 host machine to act as the central log repository and SIEM.



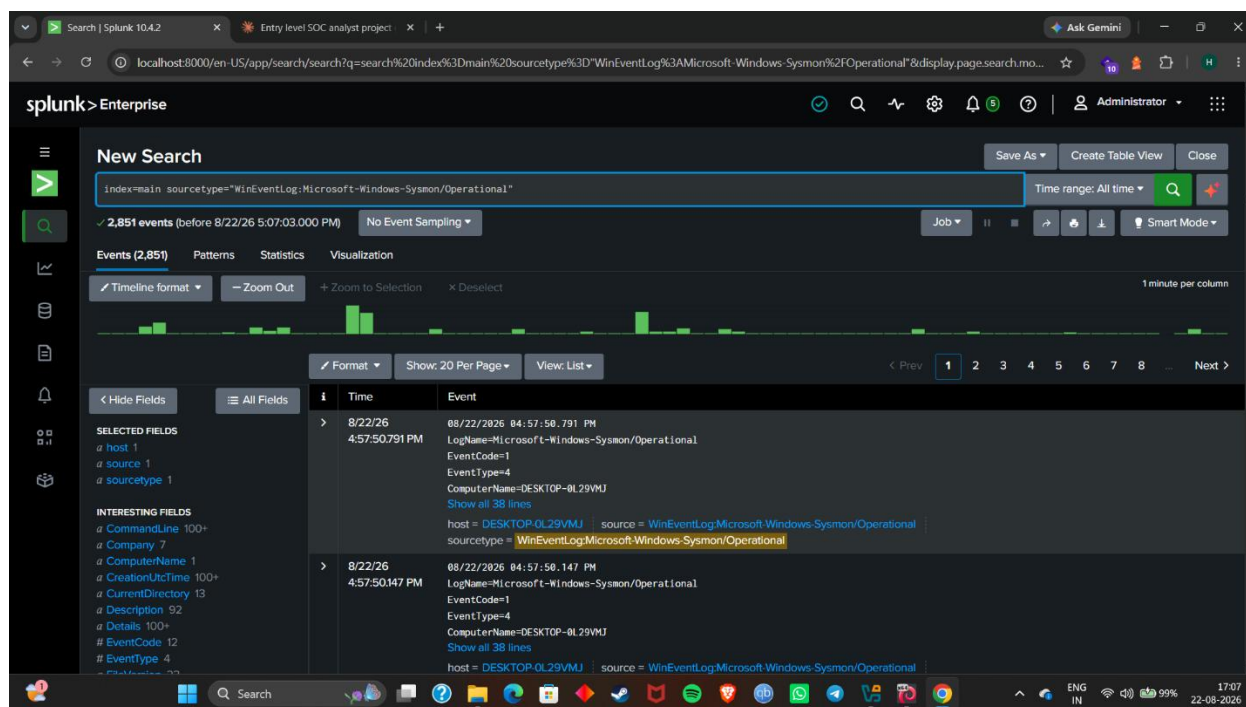
4.3 Sysmon Deployment

Sysmon was installed on the Windows 10 VM using the SwiftOnSecurity configuration, which provides detailed logging of process creation, network connections, and other system events relevant to threat detection.



4.4 Log Forwarding

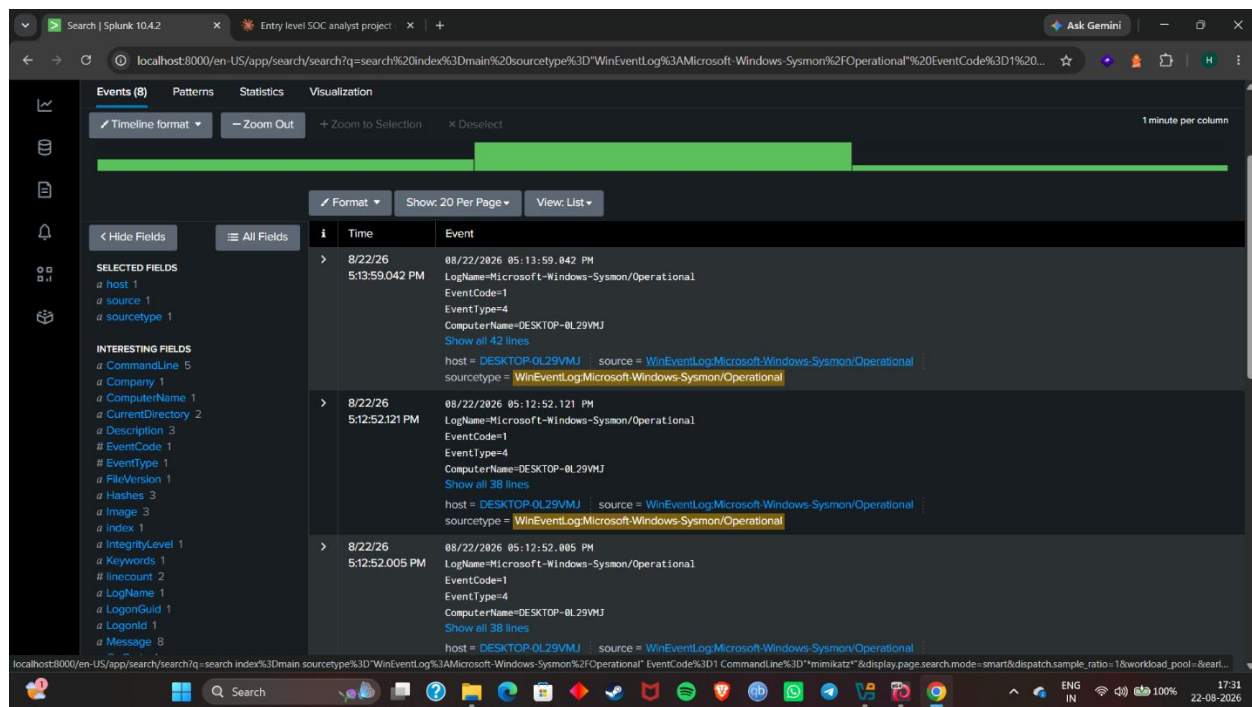
The Splunk Universal Forwarder was installed on the Windows 10 VM and configured to send both Windows Security logs and Sysmon Operational logs to the Splunk instance on the host machine over port 9997.



5. Attack Simulation

Atomic Red Team was used to execute MITRE ATT&CK technique T1059.001 (Command and Scripting Interpreter: PowerShell) on the Windows 10 VM. This technique simulates an attacker using PowerShell to execute malicious commands — one of the most common real-world attack vectors.

The simulation included multiple sub-tests, notably a Mimikatz-style credential access simulation, which mimics an attacker attempting to extract credentials from memory — a critical step in many real intrusions (mapped to T1003 – OS Credential Dumping behavior).



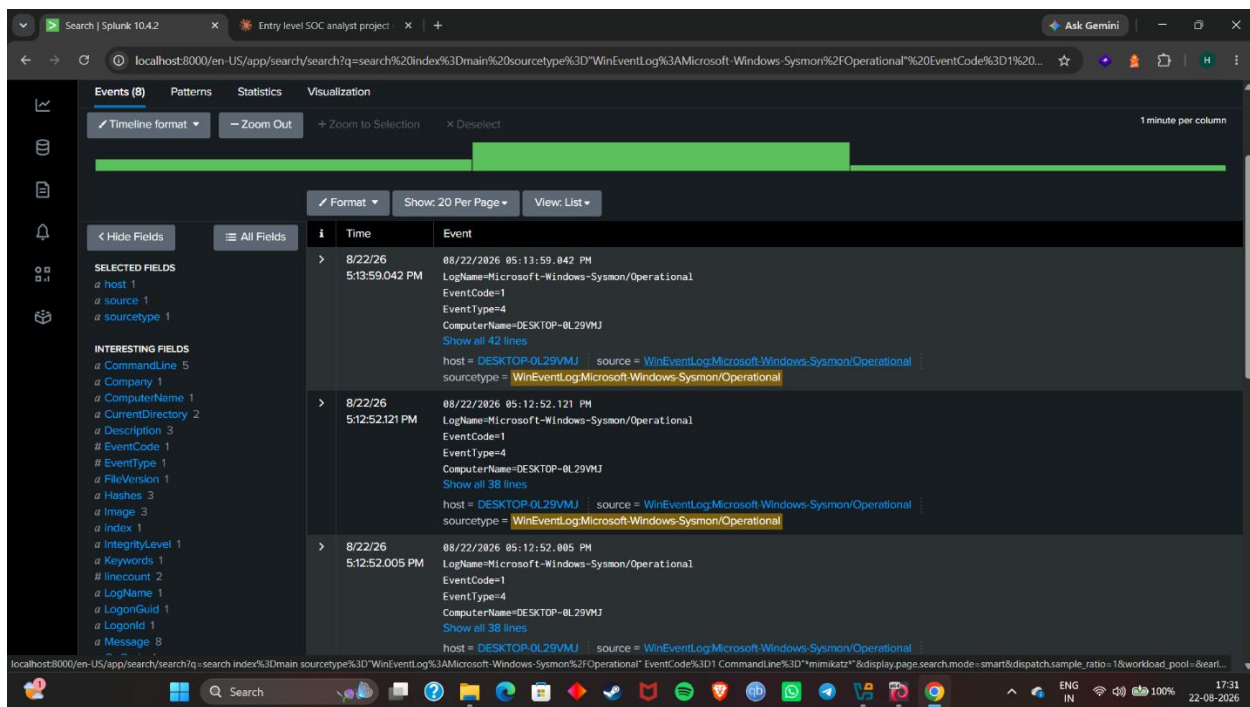
6. Detection & Analysis

6.1 Finding the Attack in Splunk

The simulated attack activity was located in Splunk using the following search:

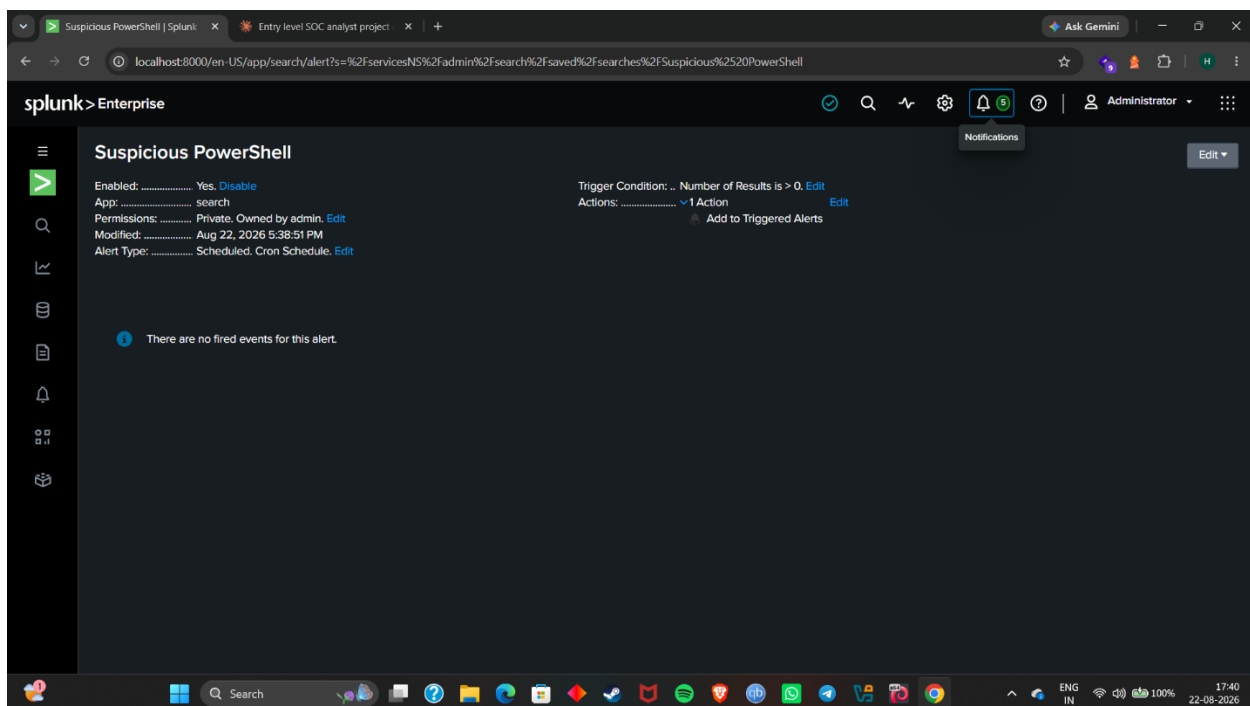
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index=main sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1
CommandLine="*mimikatz*
```

This search filters for Sysmon Event ID 1 (process creation) events where the command line contains indicators of the simulated credential access activity.



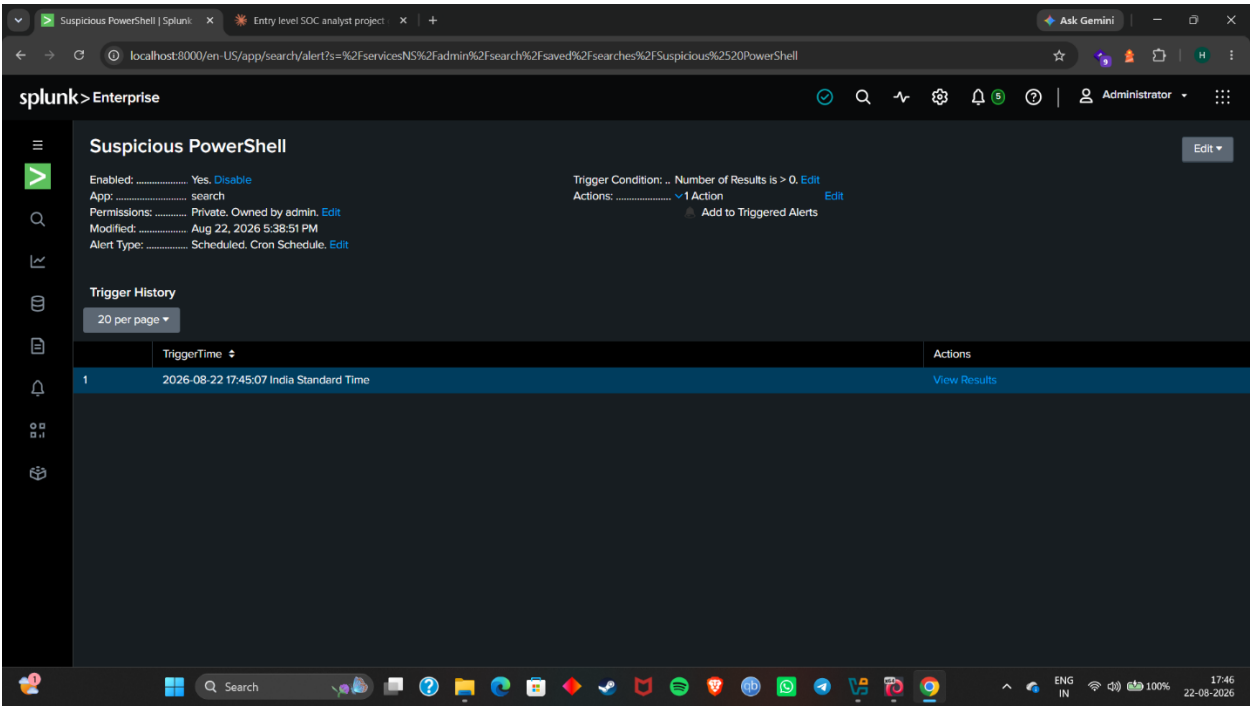
6.2 Building the Detection Alert

The search was saved as an automated Splunk alert named "Suspicious PowerShell," configured to trigger whenever the number of matching results is greater than 0. The alert runs on a scheduled cron interval and logs to Splunk's Triggered Alerts list when fired.



6.3 Alert Validation

The alert was tested by re-running the Atomic Red Team simulation and confirming that it successfully triggered within its scheduled window.



7. Incident Timeline

Event
Atomic Red Team T1059.001 executed on Windows 10 VM
Sysmon logged process creation event (EventCode=1)
Splunk Universal Forwarder shipped event to Splunk
Detection search identified the malicious CommandLine
Splunk alert "Suspicious PowerShell" triggered

8. MITRE ATT&CK Mapping

Technique ID	Technique Name	Tactic
T1059.001	Command and Scripting Interpreter: PowerShell	Execution
T1003	OS Credential Dumping (Mimikatz simulation)	Credential Access

9. Severity & Impact Assessment

In a real environment, this activity would be classified as High Severity, as it represents an attacker executing arbitrary PowerShell commands and attempting credential access — both of which are commonly early-to-mid stage indicators of a broader compromise (e.g., lateral movement or privilege escalation may follow).

10. Recommended Response Actions

If this were a real detection in a production environment, the following steps would be recommended:

- Isolate the affected host from the network immediately
- Reset credentials for any accounts active on the host at the time of the event
- Review process ancestry (parent/child processes) for further signs of compromise
- Check for persistence mechanisms (scheduled tasks, registry run keys)
- Escalate to the incident response team for full forensic review
- Review and tighten PowerShell execution policies and logging (e.g., enable PowerShell Script Block Logging)

11. Lessons Learned

- Gained hands-on experience configuring a full SIEM log pipeline from endpoint to detection
- Learned to troubleshoot real-world issues including permission errors (WinEventLog access denied), service configuration problems, and log forwarding failures
- Understood the importance of validating detections against live attack simulations rather than assuming a configuration works
- Practiced translating raw log data into a structured, MITRE ATT&CK-mapped incident report

12. Conclusion

This project demonstrates a complete, working detection pipeline: from attack simulation through log collection, detection, and automated alerting — closely mirroring the day-to-day workflow of a SOC Analyst. The hands-on troubleshooting involved (network configuration, log forwarding, permissions) further reflects the practical problem-solving skills required in real security operations roles.